SITHUM MANUSHA

SOFTWARE ENGINEERING INTERN

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ABOUT ME

A third-year Information Technology & Management undergraduate at the University of Moratuwa with strong technical competence in full-stack web development, distributed REST APIs, and relational database systems. Experienced in building modern enterprise web applications using Java (Spring Boot 3), Next.js, React, NestJS, and PostgreSQL, accompanied by automated CI/CD pipelines, Docker containerization, and real-time offline sync architectures. Seeking a Software Engineering Internship to deliver robust, scalable, and high-performance software solutions across production-grade enterprise environments.

EDUCATION

University of Moratuwa (UOM) - Moratuwa, Sri Lanka **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

St. Joseph's Vaz College - Wennappuwa, Sri Lanka **2009 - 2022**
G.C.E. Advanced Level (2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

FEATURED PROJECTS

1. CareerFlow — Full-Stack ATS Resume Optimizer & AI Interview Platform (Individual Project) **2026**

- Built a full-stack platform with a **Java 21 Spring Boot 3** gateway, **Python 3.12 FastAPI** NLP worker, and **Next.js 14** client, containerized with **Docker Compose** and automated with **GitHub Actions CI/CD**.
- Developed an automated ATS resume scanner using **TF-IDF vectorization & Cosine Similarity** to detect missing keywords, audit format health, and rewrite bullets.
- Created an interactive 5-round AI mock interview simulator with API rate limiting (10 req/min) and **26 automated tests (JUnit 5, Pytest, Playwright)**.
- **Technologies:** Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, PostgreSQL, Docker Compose, GitHub Actions, Scikit-learn, JUnit 5, Pytest, Playwright
- **Live Demo:** careerflow-app.onrender.com | **GitHub:** github.com/SithumManusha/careerflow

2. Ryzera POS — Multi-Branch Enterprise Offline-First Retail System (Group Project) **2026**

Enterprise Offline-First Architecture & Multi-Branch Real-Time Sync

- Built an offline-first sync engine using **IndexedDB and PWA Service Workers**, enabling uninterrupted cashier billing during network outages with automated batch reconciliation to **PostgreSQL**.
- Implemented a smart conflict resolution engine (using Last-Write-Wins and Additive Stock Merging), centralized runtime system configuration, and real-time terminal latency tracking via Socket.io WebSockets.
- Developed frontend screens in Next.js 16, integrated NestJS REST APIs, and performed functional API testing, boundary checks, and Role-Based Access Control (RBAC) verification using Postman.
- **Technologies:** Next.js 16, NestJS, TypeScript, PostgreSQL, Prisma ORM, IndexedDB, Socket.io, PWA, Postman, Jira, Git
- **GitHub:** github.com/SithumManusha/Ryzera-POS

3. Ledgerly — Personal Finance & Shared Expense Platform (Individual Project) **2026**

- Built an AI-assisted financial tracker that monitors monthly expenses, calculates category budget health scores, and alerts users when spending limits are reached.
- Developed an interactive balance projector using **Recharts** to visualize 6-month savings trends based on customizable monthly contributions and expense adjustments.
- Implemented group expense splitting across 4 distribution methods with automated PDF statement exports, backed by a type-safe stack (**React, TypeScript, Node.js, PostgreSQL**) and **60+ automated unit tests (Vitest)**.
- **Technologies:** React, TypeScript, Node.js, Express, PostgreSQL, Drizzle ORM, Recharts, Vitest, Render
- **Live Demo:** ledgerly-mbcd.onrender.com | **GitHub:** github.com/SithumManusha/Ledgerly

4. Automated Robotic Sorting System — Microcontroller Conveyor Platform (Group Project) **2025**

Dual-Microcontroller Mechatronics & RFID Distribution

- Constructed an automated conveyor sorting prototype equipped with a 4-DOF robotic arm, ultrasonic sensors, an RFID reader, and 3 sorting compartments.
- Engineered a two-finger mechanical gripper driven by an Arduino and stepper motor, calibrating motor step counts to achieve 98% pick-and-place precision across 3 RFID object categories.
- Tested sensor signal timings and synchronized robotic arm pick-and-place routines with continuous conveyor belt movement.
- **Technologies:** Arduino Mega/Uno (C++), Stepper Motors (NEMA 17, 28BYJ-48), ULN2003, Servo Motors, Integration Testing
- **GitHub:** github.com/SithumManusha/robotic-sorting-system

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C, SQL
- **Front-end Development:** React, Next.js 14/16, TypeScript, Tailwind CSS, State Management
- **Back-end Development:** Spring Boot 3, FastAPI, Node.js, Express.js, NestJS, RESTful APIs
- **Databases & ORM:** PostgreSQL, MySQL, Redis, Spring Data JPA, Hibernate, Prisma ORM
- **Testing & Quality:** JUnit 5, Pytest, Playwright (E2E), Vitest, Postman API Testing, Jira
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub Actions (CI/CD), Postman, Linux / Bash, Swagger / OpenAPI, Render Cloud

CORE COMPETENCIES & ARCHITECTURE

Microservices Architecture, RESTful API Design, Agile / Scrum, Object-Oriented Design (SOLID), Clean Code, Automated Testing, Offline-First Sync

CERTIFICATIONS

- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS)
- Software Testing with Generative AI – Simplilearn
- Introduction to Artificial Intelligence – Simplilearn
- Introduction to Cybersecurity – Cisco Networking Academy

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member – FIT Moments (Faculty Media Body) | Leo Club of UoM | Legal Youth Community Sri Lanka

LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

REFERENCES

Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D. (USA)),
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